

Remotes, PRs & submodules

Local Git is only half the story

Push, review, embed

- Add a remote with a URL
- Reviewers read the diff, you push more commits to the same branch if needed
- A submodule records a gitlink, the path plus an exact commit, not a casual copy of files
- After cloning a repo with submodules, you must initialize them or those folders stay empty

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Remotes, PRs & submodules

Concept quiz (22 questions), then a live GitHub checklist — clone, branch, Make, push, open a PR, and init a submodule. Uses the sandbox repos under platform/sandbox/ once published.

Practice: <https://github.com/universal-verification-methodology/unix-git-practice>

Shared IP: <https://github.com/universal-verification-methodology/unix-git-shared-ip>

SSH: `git@github.com:universal-verification-methodology/unix-git-practice.git`

Concept quiz 0 / 22 cleared

What is origin?: A remote named origin is usually...

a local branch only

the default remote pointing at the hosted repo

a commit hash

a submodule path

Show hint

Check

Next

Idle

What is origin?

Clone + submodule

First push

PR base

Empty submodule

Writable copy

Pre-push

SSH URL shape

lab1 branch

Stage sources

Submodule is

.env

Make in sandbox

gh pr

fetch vs pull

Upstream

Compare URL

After clone

shared-ip role

Why real GitHub?

Commit message

Force push

0 / 8 live steps checked · quiz separate above

1. Clone the practice repo (with submodule)

Prefer "Use this template" for a writable copy, or clone the org template. Always init submodules.

```
# option A — your own copy from the template (GitHub UI: Use this template)
# option B — clone
git clone --recurse-submodules https://github.com/universal-verification-methodology/unix-git-practice.git unix-
```

Remotes lab — quiz and live checklist

learn_git — remotes

Real Git practice

```
wsl - module15-remotes/examples/team_workflow

# Track A - real Unix
# real shell session (Track A)
# practice sandbox: /tmp/git-remotes-ObbWLg

$ git init

$ echo "practice v1" > README.md

$ git add README.md

$ git commit -m "init"

$ git branch -M main

$ git init --bare ../origin.git

$ git remote add origin ../origin.git

$ git push -u origin main

$ git checkout -b lab1

$ echo "lab1 change" >> README.md

$ git add README.md

$ git commit -m "lab1: update README"

$ git push -u origin lab1

$ git remote -v
origin    /tmp/git-remotes-ObbWLg/origin.git (fetch)
origin    /tmp/git-remotes-ObbWLg/origin.git (push)

$ git submodule add ../shared-lib external/shared-ip

$ git submodule status
849c484fba19376bdcf1d8dc322f9212ee618202 external/shared-ip (heads/master)
```

Real shell — remote, push, submodule

Real Git practice — try these

```
# git init — create a new repository here  
git init
```

```
# echo ... > README.md — first commit on default branch  
echo "practice v1" > README.md
```

```
git add README.md
```

```
git commit -m "init"
```

```
# git branch -M main — name the default branch main  
git branch -M main
```

```
# git init --bare ../origin.git — bare remote (simulates GitHub)  
git init --bare ../origin.git
```

Pitfalls to watch

- Do not force-push shared default branches like main
- After clone, run submodule init if embedded folders are empty
- A pull request is for review and visibility
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, clear a few quiz items and read the live GitHub steps end to end
- On real Git, practice push with upstream and inspect a submodule status line
- When you are ready